BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Logic Gates: AND, OR and NOT

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What must be true before binary search is used?

2. What does software copyright restrict?
   A) Using any keyboard
   B) Learning to code
   C) Saving your own work
   D) Copying without permission

3. "Proprietary software always gives every user unrestricted source-code modification rights." Correct this. Explain why.

4. A ticket program follows the language's rules but adds 5 instead of multiplying by 5. "Logic errors always stop translation." Correct this. How should the error be found?

Today's objectives

Recognise gate symbols and explain each gate. Predict outputs for specified input values.

Key vocabulary - Read, cover, recall

AND
Output 1 only when both inputs are 1.

OR
Output 1 when either or both inputs are 1.

NOT
Reverses one Boolean input.

Gate
Electronic circuit implementing a Boolean rule.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Binary logic
Logic gates work with Boolean input and output values 0 and 1.

AND gate
AND outputs 1 only if both inputs are 1. Its symbol has a flat input edge and rounded output edge.

OR gate
OR outputs 1 if either or both inputs are 1; it outputs 0 only for 0,0.

NOT gate
NOT has one input and outputs the opposite value; its symbol has a small inversion bubble.

I Do - Worked model

An alarm sounds only if a door is open (A=1) AND the system is armed (B=1). Explain the outputs for A=1,B=1 and A=1,B=0. [4 marks]

- This is an AND rule because both conditions must be true [1].
- With 1,1 the output is 1 [1], so the alarm sounds.
- With 1,0 the output is 0 [1] because both are not true [1].
- Show each input pair alongside its output before revealing.

We Do - Guided practice

1. A buzzer sounds if either A or B is on. Give OR outputs for A,B = 00,01,10,11. [4 marks]

2. A safety indicator is NOT A. Give output for A=0 and A=1. [2 marks]

3. A fan needs power A and permission B. Explain why AND is suitable and state outputs for 1,0 and 1,1. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the Boolean test so both security checks must be true.

```
signedIn = true
hasPermission = true
if ... then
print("Allow")
endif
```

You Do - Independent exam questions

Q2. State the output of an AND gate given A=1 and B=0. [1 mark]

Q3. Give OR outputs for input pairs 00, 01, 10 and 11 in that order. [4 marks]

Q4. Explain AND, OR and NOT with TWO different inputs or examples for each. [6 marks]

Exit Ticket - True or False?

An OR gate outputs 1 when both its inputs are 1.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. The list must be sorted. This lets the search safely rule out half the remaining items.

2. D - Copying without permission

3. Source code is normally controlled by its owner and users receive specified usage rights.

4. A logic error gives the wrong result even when the code follows the language's rules. Fix syntax errors first, then trace and test the logic.

I Do - Explain the model

This is an AND rule because both conditions must be true [1].

With 1,1 the output is 1 [1], so the alarm sounds.

With 1,0 the output is 0 [1] because both are not true [1].

Show each input pair alongside its output before revealing.

We Do - Model answers

1. 0,1,1,1 in the given order, one mark for each correct output.

2. For A=0 output 1 [1]; for A=1 output 0 [1].

3. Both inputs must be high [1], so AND is suitable [1]. 1 AND 0=0 [1]; 1 AND 1=1 [1].

Code First - Answer

```
signedIn = true
hasPermission = true
if signedIn AND hasPermission then
print("Allow")
endif
```

Expected output: Allow

You Do - Self-mark

Q2: 0 [1].

Q3: 0 [1], 1 [1], 1 [1], 1 [1].

Q4: AND 1,1 -> 1 [1] and 1,0 -> 0 [1]. OR 0,1 -> 1 [1] and 0,0 -> 0 [1]. NOT input 0 -> 1 [1] and input 1 -> 0 [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.